DATA & FEATURE REQUIREMENTS · VERSION 1.0 · OCTOBER 2026

What NeoStay needs *from the hospital*

A request, field by field: what the platform already has, what must come from the hospital's own records, how much of it is needed and why, how it should be prepared and transferred, and what the hospital receives in return.

For discussion with hospital management

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PURPOSE
Agree the data that turns a working prototype into a validated service

AUDIENCE
Hospital management, informatics, information governance

DATA SOUGHT
Pseudonymised, aggregate where possible, no direct identifiers

DECISION NEEDED
Named data owner and approval to start the governance route

00 Contents and the ask in brief

One page for a busy reader; the detail follows.

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The ask in three sentences. NeoStay works today on aggregated statistics and the unit's own acuity policy. To make its forecasts accurate for *this* hospital — and to prove they are — it needs a pseudonymised extract of past neonatal admissions, the unit's census and staffing history, and the local nursing policy. No names, addresses or record numbers are requested, and no patient-level data is stored in the product itself.

8
DATASETS REQUESTED

5 yrs
PREFERRED HISTORY (2 MINIMUM)

0
DIRECT IDENTIFIERS NEEDED

01 The ask, in one page

What we need, from whom, and by when.

TABLE 1 — SUMMARY OF THE REQUEST

Dataset	Granularity	Minimum	Preferred	Held by
D1 · Infant admissions	One row per admission	2 years	5–10 years	Neonatal database / EHR
D2 · Clinical course	One row per infant per day	2 years	5 years	Neonatal database
D3 · Acuity and level of care	One row per infant per shift	3 months	12 months	Nursing records / acuity sheets
D4 · Unit census and flow	Hourly or per shift	12 months	2–3 years	Bed management / EHR
D5 · Nursing establishment	One row per grade per month	12 months	2 years	Workforce / rostering system
D6 · Actual rosters	One row per nurse per shift	6 months	12 months	Rostering system
D7 · Local policy	Documents	Current version	With change history	Nursing leadership
D8 · Branch metadata	One row per unit	Current	With historical changes	Service management

Start small if that is easier. D7 and D8 are documents and can be agreed in a single meeting; they unlock configuration immediately. D3 and D4 unlock the staffing work. D1, D2 and D6 unlock the predictive models. Each dataset is useful on its own — nothing has to wait for everything.

What is explicitly not requested

- Names, addresses, dates of birth, hospital or national identifiers of babies, mothers or staff.
- Free-text clinical notes, images, or anything from which an individual could be recognised.
- Payroll, pay bands as monetary values, individual performance records, or sickness reasons.
- Any data about babies who did not receive care in the participating units.

02 What NeoStay already holds

So the request is limited to what is genuinely missing.

TABLE 2 — DATA ALREADY IN THE PLATFORM

Source	Contents	Granularity	Used for
Aggregated outcome tables 2016–2026, one centre	For 11 birth-weight bands and 18 gestational-age groups: number of infants, median, Q1 and Q3 length of stay; the same split by discharge outcome (home, transfer, died). 712 infants in total.	Group statistics, no individual rows	Today's length-of-stay and outcome estimates
Acuity tool and levels of care	11 body systems with 108 selectable findings across four nurse:patient ratio columns, and 64 level-of-care criteria (45 general, 19 hyperbilirubinemia).	Policy document, transcribed	Classifying each cot; setting ratios and minimum nurse level
Synthetic unit activity	A deterministic simulation of 709 infants and 42 nurses over a decade, anchored to the real medians and outcome shares.	Hourly, synthetic	Demonstration of the timeline only — never used for prediction

The limitation this request addresses. The aggregated tables cannot support a fitted model: they contain no individual records, no clinical course, and nothing about staffing. They also describe one centre over a decade, so they cannot tell another branch what to expect. Everything requested below exists to remove those two limits.

03 Feature inventory: have, need, why

Every variable the models need, and where it stands today.

TABLE 3 — FEATURE INVENTORY

Feature group	In NeoStay today	Needed from the hospital	Priority
Birth characteristics	Birth weight and gestational age, as band statistics only	Per infant: exact weight, gestation in weeks + days, sex, singleton or multiple, inborn or outborn, antenatal steroids, mode of delivery, Apgar at 1 and 5 minutes, admission temperature	Must
Admission severity	A proxy: a severity score computed from weight and gestational age, raised by condition and respiratory support	First-hour blood gas (pH, base excess), FiO_2, oxygenation index, blood pressure support, CRIB-II or SNAPPE-II if recorded, major congenital anomaly	Must
Clinical course	Nothing — the current model assumes a fixed three-phase pathway	By day: respiratory support type, surfactant, inotropes, sepsis episodes, NEC, IVH grade, PDA treatment, ROP treatment, BPD at 36 weeks, surgery, cooling	Must
Feeding and growth	Nothing	First enteral feed day, full feeds day, parenteral nutrition days, central line days, weights at 7/14/28 days and discharge	Should
Acuity per shift	The classification rules, and whatever a charge nurse enters live	Historical record: level of care and ratio applied per infant per shift, with the findings that justified it	Must
Unit census and flow	Synthetic only	Real census by hour or shift; admissions, discharges, transfers in and out, deaths with timestamps; cot capacity and closures; refused admissions	Must
Workforce	Four modelled levels with assumed weightings; the user types in how many are available	Establishment by grade; actual staffing per shift by grade; bank and agency use; vacancy and sickness rates; qualified-in-speciality proportion	Must
Local policy	One hospital's acuity tool and an assumed mapping to four nurse levels	This hospital's grade ladder, ratio policy, supervision rules, escalation policy, and the levels of care in use at each branch	Must
Outcomes	Discharge outcome shares per weight band	Per infant: discharge date and destination, death date, transfer destination and reason, readmission within 28 days	Must

Feature group	In NeoStay today	Needed from the hospital	Priority
Calendar and context	Nothing	Public holidays, known seasonal pressures, planned cot closures, service changes with dates	Should
Quality indicators	Nothing	Unit-level indicators already reported nationally (mortality, NEC, IVH, BPD, ROP, breast-milk rates), so the models can be checked against known benchmarks	Could

04 Datasets requested, field by field

Hand these tables to informatics: they are an extract specification.

D1 · Infant admissions — one row per admission

Field	Type / example	Why it is needed	Priority
`study_id`	Text, e.g. `NS-000142`	Pseudonymous key. Links D1, D2 and D3. The look-up table stays with the hospital.	Must
`branch_id`	Text, e.g. `BR-01`	Which unit; enables per-branch validation.	Must
`admission_datetime`	ISO 8601, to the hour	Time origin for length of stay; drives census and seasonality.	Must
`discharge_datetime`	ISO 8601, blank if still inpatient	Outcome time; blanks are handled as censored, not discarded.	Must
`discharge_destination`	`home | transfer | died | other`	The competing outcomes.	Must
`birth_weight_g`	Integer grams	Strongest single predictor.	Must
`gestational_age_weeks`, `_days`	Integer, e.g. 25 + 3	Second strongest; days matter at the extremes.	Must
`sex`	`male | female | indeterminate`	Established prognostic factor.	Must
`multiple_birth`, `birth_order`	Boolean, integer	Twins and triplets behave differently and arrive together, which matters for staffing.	Should
`inborn`	Boolean	Outborn babies arrive sicker and later.	Must
`antenatal_steroids`	`none | partial | complete`	Changes respiratory course substantially.	Must
`delivery_mode`	`vaginal | caesarean`	Context for early course.	Should
`apgar_1`, `apgar_5`	Integer 0–10	Early severity.	Must
`admission_temperature_c`	Decimal	Known association with outcome; quality indicator.	Should
`ph_first_hour`, `base_excess_first_hour`	Decimal	Components of CRIB-II / SNAPPE-II.	Should
`crib_ii` or `snappe_ii`	Integer	If already computed, a strong single severity feature.	Should
`major_anomaly`	Boolean + SNOMED CT code	Changes pathway entirely; often excluded from comparisons.	Must

Field	Type / example	Why it is needed	Priority
`readmission_28d`	Boolean	Quality signal, and affects true length of care.	Could

D2 · Clinical course — one row per infant per day

Field	Type / example	Why it is needed	Priority
`study_id`, `day_of_stay`	Text, integer	Links to D1; builds the daily trajectory.	Must
`respiratory_support`	`ventilation | hfov | nippv | cpap | high_flow | cannula | room_air`	Primary driver of nursing intensity and of stay.	Must
`fio2_max`	Decimal 0.21–1.0	Severity within a support type.	Should
`surfactant_doses`	Integer	Early respiratory severity.	Should
`inotropes`, `sedation`	Boolean	Instability; 1:1 criteria in many policies.	Must
`central_line`, `line_type`	Boolean, text	Nursing workload and infection risk.	Must
`parenteral_nutrition`, `enteral_feeds`	Boolean, `none | trophic | partial | full`	Marks the step-down pathway.	Must
`phototherapy`, `exchange_transfusion`	Boolean	Directly in the acuity tool.	Should
`sepsis_suspected`, `sepsis_confirmed`	Boolean	Common cause of a sudden rise in acuity.	Must
`nec_stage`, `ivh_grade`, `pda_treatment`, `rop_treatment`, `surgery`, `cooling`	Coded	Major events that lengthen stay and raise acuity.	Must
`weight_g`	Integer	Growth trajectory; discharge readiness.	Should

D3 · Acuity and level of care — one row per infant per shift

Field	Type / example	Why it is needed	Priority
`study_id`, `shift_start`	Text, ISO 8601	Links the clinical picture to the nursing requirement.	Must
`level_of_care`	`N1 | N2 | N3 | N4` or local equivalent	The label the acuity model learns to predict.	Must
`ratio_applied`	`1:1 | 1:2 | 1:3 | 1:4`	What the unit actually did, versus what policy implies.	Must
`one_to_one_reason`	Coded or short text	Validates the footnoted 1:1 criteria.	Should
`findings`	List of acuity-tool items	Allows the classifier to be checked against real practice.	Should
`nurse_grade_assigned`	Local grade	Whether competence matched acuity in practice.	Should

D4 · Unit census and flow — hourly or per shift

Field	Type / example	Why it is needed	Priority
`branch_id`, `timestamp`	Text, ISO 8601	The demand time series.	Must
`census_total`, `census_by_level`	Integers	Demand, and its acuity composition.	Must
`admissions`, `discharges`, `transfers_in`, `transfers_out`, `deaths`	Integers	Arrival and departure processes for forecasting and simulation.	Must
`cots_open`, `cots_closed_reason`	Integer, text	Capacity actually available.	Must
`refused_admissions`	Integer	Suppressed demand — invisible in census alone.	Should

D5 · Establishment · D6 · Actual rosters

Field	Type / example	Why it is needed	Priority
`branch_id`, `month`, `grade`, `wte_funded`, `wte_in_post`	Text, date, decimal	What the unit is funded for versus what it has.	Must
`qis_proportion`	Decimal	Qualified in speciality: the skill-mix benchmark many units already report.	Should
`staff_pseudo_id`, `shift_start`, `grade`, `role`	Text, ISO 8601	Who was actually on, by grade and role (bedside, charge, float). No names.	Must
`bank_or_agency`, `overtime`	Boolean	Cost and sustainability of the current approach.	Should
`absence`	Boolean (no reason)	Short-notice gaps, for robustness planning. Reasons are not requested.	Should

D7 · Local policy · D8 · Branch metadata

Item	Form	Why it is needed	Priority
Nurse grade ladder and its mapping to four competency levels	One page, signed by the nursing lead	Makes the tool speak the hospital's language rather than Benner's.	Must
Ratio policy, including local 1:1 criteria	Current policy document	Replaces the assumptions currently built in.	Must
Supervision rules (novice cap, preceptorship)	Policy or written confirmation	Drives the skill-mix warnings.	Must
Acuity tool in use at each branch	Document or spreadsheet	Branches may classify differently; the classifier reads this.	Must
Branch list: designated level, cot capacity by level, shift pattern	Table	Configuration, and the denominator for every comparison.	Must

05 How much data, and what each amount buys

Volumes are driven by statistics, not by appetite.

The rule behind the numbers. A prediction model needs roughly 10 to 20 outcome events for every variable it considers, plus a separate, untouched period to test on. With about 25 candidate variables (Section 3), that means several hundred completed admissions as a floor, and one to two thousand to produce stable estimates with subgroup results. Time-series forecasting is governed differently: it needs enough repetitions of the seasonal cycle, which for a weekly and annual pattern means at least one, and preferably two to three, full years.

TABLE 4 — VOLUME, AND WHAT IT ENABLES

Dataset	Minimum viable	What that allows	Preferred	What the extra buys
D1 + D2 admissions and course	~300 completed admissions (about 2 years in a 150-admission unit)	A first length-of-stay model with a handful of variables; internal validation only; wide intervals.	1,000–2,000 admissions (5–10 years, or 2 years pooled across branches)	Competing-risks modelling, subgroup results by weight band, external validation, stable intervals.
D3 acuity per shift	3 months (~2,700 infant-shifts in a 15-cot unit)	Check the classifier against real practice; estimate transition rates between levels.	12 months (~11,000 infant-shifts)	Seasonal variation, reliable next-shift prediction, per-branch comparison.
D4 census and flow	12 months hourly (~8,800 rows per unit)	Weekday and seasonal pattern; a forecast that beats the naive baseline.	2–3 years	Confident seasonality, rare-event behaviour (surges, closures), simulation input.
D5 + D6 establishment and rosters	6 months of rosters	Compare recommended with actual staffing; calibrate the skill-mix rules.	12–24 months	Robust establishment case; fairness and continuity objectives for the roster optimiser.
D7 + D8 policy and branches	Current versions	Configure each branch correctly — needed before any pilot.	With change history	Explains shifts in practice over the period being modelled.

If the data is thinner than the minimum

- **Pool across branches** with a hierarchical model, so a small unit borrows strength without losing its own signal.
- **Reduce the variable count** deliberately, using clinical judgement rather than automated selection, and report that the model is deliberately simple.
- **Stay with the deterministic method** and publish its error. A transparent, honest baseline is better than an over-fitted model.

What "enough" looks like in practice. For a unit admitting about 150 very low birth weight infants a year: five years of D1 and D2, one year of D3 and D6, and two years of D4 would support everything described in the dossier, including per-branch validation.

06 Data quality expectations

What makes an extract usable, stated before the work starts.

TABLE 5 — QUALITY CRITERIA

Criterion	Expectation	If it cannot be met
Completeness of key fields	Birth weight, gestation, admission and discharge dates, outcome: above 95% complete	We report results for the complete subset and quantify what is missing.
Completeness of secondary fields	Above 70%	Multiple imputation, with a sensitivity analysis reported.
Consistency	Discharge after admission; weights within plausible ranges; gestation 22–42 weeks	Rows flagged and returned to the hospital rather than silently corrected.
Coding	SNOMED CT or ICD-10 where codes exist; a documented local list otherwise	We map the local list once, and keep the mapping under version control.
Timestamps	Local time with a stated time zone; hour precision is enough	Shift-level aggregation instead of hourly.
Stability	A note of any service change, policy change or system migration during the period	Models include the change as a break point rather than learning a false trend.
Reproducibility	The extract query saved and re-runnable	Annual revalidation becomes manual and expensive.

07 Format, pseudonymisation and transfer

Practical instructions for whoever runs the extract.

File format

One UTF-8 CSV (or Parquet) per dataset, with a header row and the field names used in Section 4. One row per record; no merged cells; no colour coding; no totals rows. A short data dictionary alongside, naming every coded value.

Pseudonymisation

Replace every patient and staff identifier with a study identifier (`NS-000001` , `ST-0001`). The look-up table stays inside the hospital, held by the data owner. Dates may be shifted by a constant per infant if required, provided intervals are preserved.

Transfer

Through the hospital's approved secure mechanism — managed file transfer, a secure shared workspace, or encrypted media. Not by personal email, consumer cloud storage or messaging apps. A checksum for each file, verified on receipt.

Handling once received

Stored encrypted, inside the agreed boundary, accessible only to named analysts, never copied to personal devices, and destroyed at the end of the agreed retention period with written confirmation.

Example header rows

```
# D1_admissions.csv
study_id,branch_id,admission_datetime,discharge_datetime,discharge_destination,birth_weight_g,
gestational_age_weeks,gestational_age_days,sex,multiple_birth,inborn,antenatal_steroids,
delivery_mode,apgar_1,apgar_5,admission_temperature_c,major_anomaly

# D4_census.csv
branch_id,timestamp,census_total,census_n1,census_n2,census_n3,census_n4,admissions,discharges,
transfers_in,transfers_out,deaths,cots_open,refused_admissions
```

08 Governance, approvals and responsibilities

The route that must be completed before any patient-level extract moves.

TABLE 6 — APPROVALS AND OWNERS

Step	What it involves	Owner	Before
Classify the work	Service evaluation or research? Determines whether ethics approval is required.	Clinical Lead with research office	Everything else
Name a data owner	A single accountable person per branch for the extract and the identifier look-up.	Hospital management	Extract
Data protection impact assessment	Document flows, minimisation, risks, retention.	Information governance	Extract
Data sharing agreement	Purpose, scope, security, retention, destruction, named recipients.	Hospital legal / IG	Extract
Caldicott or equivalent sign-off	Confirms the minimum necessary data is being shared.	Caldicott Guardian	Extract
Security assurance	Where the data will live, who can reach it, how it is encrypted and destroyed.	Technical Lead with hospital IT	Extract
Publication agreement	Who approves any publication, and how branches are described.	Both leads	Analysis

No data moves before this is complete. The technical team does not request, receive or store any patient-level data until the data owner is named and the agreement is signed. Configuration work (D7, D8) and the pilot can proceed in parallel, because they involve no patient data at all.

09 Timeline, and what the hospital receives back

A fair exchange, with dates.

TABLE 7 — INDICATIVE TIMELINE FROM APPROVAL

Weeks	Hospital provides	Project delivers
0–2	Policy documents (D7), branch list (D8), pilot unit and champions	Branch configuration; training session; quick-reference card
2–6	Governance route started; census and roster history (D4, D6)	Parallel-running report: recommended versus actual staffing, with the gaps named
6–12	Pseudonymised clinical extract (D1, D2, D3)	Data quality report; first length-of-stay and demand models with honest error figures
12–20	Questions answered; clinical review of error cases	Validation report to TRIPOD+AI; model cards; a decision recommendation — adopt, adapt or keep the simpler method
20+	Decision on wider rollout	Per-branch configuration, dashboards, annual revalidation cycle

What the hospital gets, regardless of what the models show

- A written data quality report on its own neonatal records — often valuable in itself.
- A description of its own demand: census by hour, day and season, and the acuity mix behind it.
- A comparison of recommended and actual staffing, suitable for an establishment review.
- Per-branch validation figures, published in full, including where the method performs poorly.
- The configured application, running inside the hospital's own boundary.

10 Sign-off and checklists

To be completed at the meeting, so nothing stalls afterwards.

Decisions to record today

- ☐ Pilot unit agreed, with two named champions
- ☐ Data owner named, per branch
- ☐ Governance route started (classification, DPIA, sharing agreement)
- ☐ Policy documents identified and owner named (D7, D8)
- ☐ Extract feasibility confirmed with informatics, with a rough date
- ☐ Hosting model chosen: hospital server, hospital cloud, or managed platform
- ☐ Date set for the next review

Informatics checklist for the extract

- ☐ Field names and coded values follow Section 4
- ☐ Study identifiers replace all patient and staff identifiers; look-up retained by the hospital
- ☐ Period and branches agreed in writing
- ☐ Still-inpatient records included, with a blank discharge date rather than being dropped
- ☐ Data dictionary and extract query saved with the files
- ☐ Checksums generated; transfer through the approved mechanism
- ☐ Known data breaks (system migrations, policy changes) noted with dates

TABLE 8 — SIGN-OFF

Role	Name	Date	Signature
Clinical Lead (project)			
Hospital management sponsor			
Nursing leadership			
Information governance			
Informatics / data owner			

NeoStay — Data and Feature Requirements, version 1.0. Prepared for discussion with hospital management by Waseem Altaf, MD, Project and Clinical Lead, Neonatologist. Specification by Abdul Razak, PhD, Technical Lead, Lead Developer and Applied AI/Data Science Lead. No patient-identifiable data is requested or stored. Decision support only — not for clinical use. © 2026 Waseem Altaf, MD. Source: `docs/architecture/neostay-data-requirements.html`.